System context

English version of ../../architecture/system-context.md.

What this is

A web application for assessing service-renewal risk and the revenue at risk. The risk score is an **explainable weighted heuristic**, not a churn probability: the dataset has no labelled outcomes. That is a product limitation, not an implementation defect.

Context diagram

Two databases — a deliberate decision

- **PostgreSQL** — domain data (accounts, services, transactions, aggregates).
- **SQLite** — authentication only (`AUTH_DATABASE`).

The point: the authentication module can be used and tested on its own without PostgreSQL (`tests/test_auth.py` runs without the analytics database). The price: two backup mechanisms, both described in operations/deployment.md.

Supported scope

Declared: PostgreSQL 16, Python 3.12, Linux + Docker Compose, Chromium-compatible browsers. Not declared: other DBMS, running the analytics part without PostgreSQL, scaling to several `api` replicas without external state (the snapshot and the cache live in the process).

Failure modes of external systems

Dependency	Mandatory	When missing
PostgreSQL	yes	the application serves no analytics
LLM (Ollama and others)	no	AI features are off, calculations work
SMTP	no	e-mail is not sent
Telegram	no	the integration is off